

PRODUCT KNOWLEDGE • TRAINING DECK FOR DEALERS AND FIELD STAFF

Vital Urea

Sulfur Coated Urea • N 32% min • S 13% min

Why Pakistani soil needs sulfur with its nitrogen, what is in the bag, how it releases, and how to apply it. Every figure carries its source.

PS 217-2023

Licence CM/L-4126/2025

25 kg bag

Patent No. 144684



Alkaline, low in organic matter, and never tested for sulfur.

From the Punjab soil testing programme: 770,160 samples, 36 districts, sampled c. 2016 to 2018.

96.1%

of samples above pH 7.5

All 36 district averages are above pH 7.5. 69.1% of samples are above pH 8.

90.2%

below 0.86% organic matter

The average is 0.61%. All 36 district averages are below 0.86%.

0

of 770,160 samples tested for sulfur

Each sample was tested for 13 properties. Sulfur was not one of them.

3.68 to 1

nitrogen to phosphorus used

Against 2 to 1 recommended. The review does not report sulfur as a nutrient at all.

What to tell a farmer: his soil report has never shown sulfur because sulfur was never tested. That is different from knowing his field has enough.

Soil: Punjab soil testing programme, 36 district workbooks, 770,160 samples, as published on van.com.pk. Fertilizer ratio: NFDC Fertilizer Review 2024-25.

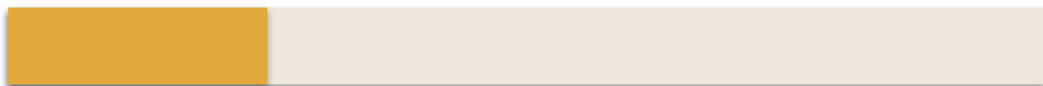
Above pH 8, part of the surface urea turns to ammonia gas and is lost.

Plain urea dissolves all at once. From that moment, every kilogram of it is exposed.

Share of applied nitrogen taken up by the crop

Pakistan

about 25%



United States

about 72%



Lassaletta et al. (2014), Environmental Research Letters 9:105011.

1

Urea dissolves

A granule of plain urea dissolves all at once in the first water.

2

Ammonia leaves

On soil above pH 8, surface urea turns to ammonia gas.

3

Nitrate moves down

What is left can wash below the roots with irrigation water.

4

What the coating changes

Nitrogen still inside a granule is not in the soil water yet, so it cannot be lost yet.

A coating does not stop volatilisation. It reduces how much nitrogen is exposed at any one moment. Placement and irrigation timing still matter.

Say sulfur in Punjab is “not measured”, and never “deficient”.

The evidence that exists points both ways. Field staff should be able to explain both.

Across the border, on the same kind of soil

50.2%

of soils in Indian Punjab deficient,
counting latent deficiency

78.2%

in Rajasthan, which borders Sindh and
south Punjab

Shukla et al. (2021), Scientific Reports, doi:10.1038/s41598-021-99040-2. Indirect evidence from a neighbouring area. It says nothing certain about a field in Pakistan.

What VAN does not claim

A 2003 study of 45 samples from Punjab's Kallar tract found sulfur adequate in most of them (Ahmad et al., 2003, Asian Journal of Plant Sciences 2(5)).

So VAN says sulfur in Punjab has not been measured at scale. VAN does not say every Punjab field is short of sulfur.

Why it still matters: every harvest removes sulfur (slide 12), soils here are low in organic matter, and national fertilizer use leans on nitrogen. Vital Urea puts sulfur beside the nitrogen with no extra field pass.

What is in a 25 kg bag of Vital Urea.

Urea granules coated with finely ground elemental sulfur, held on by a binder VAN developed and patented.

Item	What it is
Bag	25 kg
Declared analysis	32% nitrogen, 13% sulfur: at least 8.0 kg N and 3.25 kg S per bag
Sulfur	Elemental sulfur ground to 350 mesh or finer
Coating material A	Carbohydrate (starch-based) binder that holds the sulfur on the granule
Coating material B	The release-control component
On the pack	Slow Release Fertilizer
Form and colour	Prill, 1 to 3 mm (more than 90%), light to dark yellow

The patent covers the combination of the 2 coating materials. VAN describes the process as low in energy use and safe to handle, and the coating as biodegradable, leaving no residue that reacts badly with the soil, its minerals or the crop.



Analysis and registration: PS 217-2023, licence CM/L-4126/2025. Form and colour: VAN SDS-VAN-001 Rev 3, July 2026.

Registered, patented, licensed, and tested batch by batch.

The dates a dealer can quote.

2012

VAN starts work on coated products and has tested many combinations since 2012, including biological, nutrition-based and biostimulant coatings. This one was chosen from them.

2014

The standard for VAN's sulfur coated urea is approved.

2019

The product is registered, and has been ever since.

2026

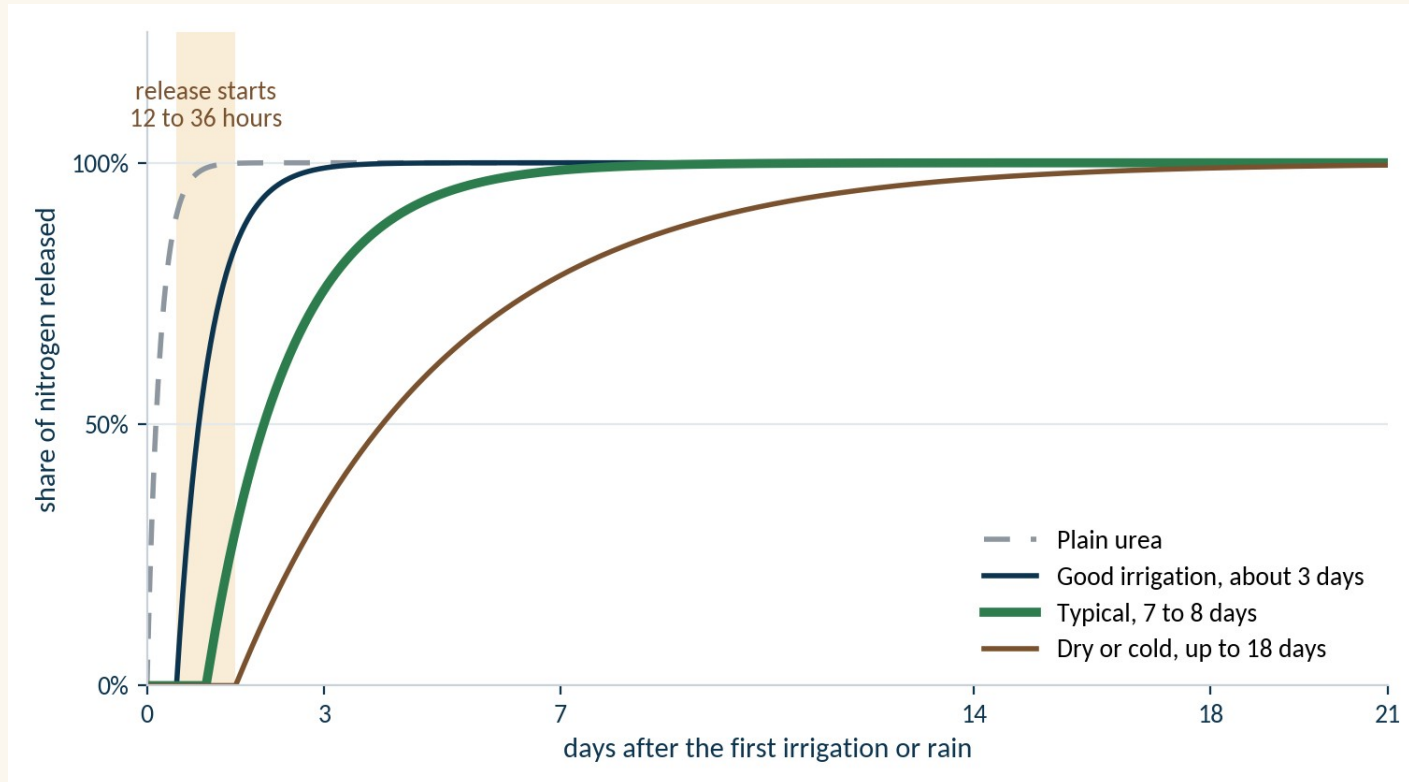
Pakistan Patent No. 144684 is completed. VAN holds the patent.

Now

Made under PSQCA standard PS 217-2023, licence CM/L-4126/2025. Every batch is released through VAN's PNAC-accredited laboratory, ISO/IEC 17025:2017, LAB 336.

Certificate on request: send the batch number printed on the bag and VAN sends the certificate of analysis for that batch.

Nothing happens until water arrives. The clock starts at the first irrigation or rain.



12 to 36 hours

after the first irrigation, release starts

about 3 days

to full release under good irrigation

7 to 8 days

to full release in typical conditions

up to 18 days

when the ground is dry or cold

What happens: water breaks down the carbohydrate binder. That lets go of the fine sulfur, which spreads into the soil around the granule, and the urea starts to dissolve. A granule crushed at spreading has lost its coating and behaves like plain urea.

VAN's own field observation of when release starts and ends, curves drawn for shape between those points. This is not a laboratory release curve.

Soil bacteria can only work on the outside of a sulfur particle.

Elemental sulfur feeds the crop only after bacteria turn it into sulfate. They do that at the surface.

350 mesh

or finer: VAN's sulfur grind

2 times

the surface when the particle size is halved, for the same kilogram of sulfur

20 μm

rapid oxidation needs particles finer than about this, well dispersed (Bremer, 2022)

3 to 4 times

faster oxidation for every 10 °C of soil warming (Janzen & Bettany, 1987)

The reaction: $2 \text{S}^0 + 3 \text{O}_2 + 2 \text{H}_2\text{O} \rightarrow 2 \text{SO}_4^{2-} + 4 \text{H}^+$. Sulfate is the form a root takes up. As it oxidises, the sulfur lowers the pH in a small zone around each granule, which may help nearby phosphorus and zinc. It does not change a soil test.

Sulfur oxidises more slowly in alkaline soil: 12% against 20.9% in an acidic soil, in a laboratory comparison (Mattiello et al., 2017). On Punjab's soils that is one more reason the sulfur needs to be fine, and in the soil early.

Without sulfur, nitrogen taken up cannot become protein.

2 of the amino acids that make up protein, cysteine and methionine, contain sulfur.

Sulfur short

Protein building slows. The nitrogen the plant has taken up builds up unused, as free amino acids such as asparagine and arginine.

Yu et al. (2018), Scientific Reports 8:2499, doi:10.1038/s41598-018-20935-8

More than 20%

rise in wheat nitrogen use efficiency when sulfur was added to sulfur-deficient soil.

Yu et al. (2021), Communications Biology 4, 945, doi:10.1038/s42003-021-02458-7

0.15 to 0.32

wheat nitrogen recovery without and with sulfur, on a poor-fertility site in Argentina.

Arata et al. (2017), Crop & Pasture Science 68(3), doi:10.1071/CP16330

These are published studies of sulfur and nitrogen. None of them is a measurement of Vital Urea, and no yield figure is claimed for Vital Urea from them.

Wheat protein holds 12 to 15 parts nitrogen for each part sulfur.

12 to 15 : 1

nitrogen to sulfur in wheat grain protein

Yu et al. (2018); Grzebisz et al. (2022)

17 : 1

above this grain N:S, with grain sulfur below 0.12%, wheat responded to sulfur

Randall, Spencer & Freney (1981), Aust. J. Agric. Res. 32(2), doi:10.1071/AR9810203

12 to 15 kg

of nitrogen that cannot become protein for each kilogram of sulfur the crop is short of

Derived: 1 kg of sulfur × 12 to 15 parts nitrogen per part sulfur. A cereal ratio; brassicas run lower.

In the field: each Vital Urea granule carries nitrogen and sulfur together, at 32 to 13 in the bag, so the 2 arrive in the same place at the same time.

Use the ratio as a guide. Randall et al. (1981) suggest grain analysis as a way to check a crop's sulfur status in a given field.

Each crop takes sulfur from the soil every season.

1 bag of Vital Urea puts 3.25 kg of sulfur on an acre. It does not replace a whole season's removal.

Crop	Sulfur taken up or removed	Published figure and source
Wheat	8.9 kg S per acre	22 kg S/ha taken up, grain and straw, by a 4.6 t/ha crop (about 46 maunds an acre) · FAO Bulletin 16 (2006)
Maize	8.5 kg S per acre	21 kg S/ha removed in grain and stover at 9.5 t/ha (about 96 maunds an acre) · FAO Bulletin 16 (2006)
Sugarcane	8.1 to 13.0 kg S per acre	20 to 32 kg S/ha · Shukla & Lal (2004), Indian Journal of Agronomy 49(1)
Cotton	4.0 kg S per acre	10 kg S/ha taken up by 2.5 t/ha of seed cotton (about 25 maunds an acre) · FAO Bulletin 16 (2006)
Rice	about 2 kg S per tonne of grain	IRRI rice fact sheet (2003)
1 bag of Vital Urea	3.25 kg S per acre	13% of 25 kg, the registered analysis

Per-acre figures are the published per-hectare figures divided by 2.471. Uptake is what the whole crop takes in; removal is what leaves the field. Both are reference values at the yields stated, not measurements from Pakistani fields.

Broadcast, drill or side-dress, at the earliest stage, and let water follow it. Never fertigate.

1

Broadcast, drill or side-dress

Broadcast it, drill it at sowing, or side-dress or band it beside the row. The granule needs soil and water around it.

2

Use it early

Apply at the earliest stage the crop plan allows. It can be the first or the second urea application. Earlier is better.

3

Count from the water

Release starts after the first irrigation or rain. Plan the application so that water follows it.

4

Never fertigate

Do not dissolve it and run it through drip or irrigation water. It is made to release slowly in the soil.

Handle with care: a granule crushed in the spreader has lost its coating and behaves like plain urea. Handle bags so the granules are not crushed.

The rate and the stage always come from the crop's own VAN plan. The next 2 slides list them, per acre.

The rate and stage from each crop's VAN plan, per acre.

Crop	Stage	Vital Urea per acre
Wheat	Land preparation	½ bag (12.5 kg)
Maize	Land preparation; germination	1 bag; 1 bag
Cotton	Germination; grand growth	1 bag; 1 bag
Sugarcane, February planting	Germination	1 bag + ½ bag in March; 1 bag + ½ bag in April (2 applications each month)
Ratoon sugarcane	Land preparation; germination	1 bag; 1 bag
Rice, basmati and hybrid	Early growth	1 bag
Potato	Land preparation; germination	1 bag; 1 bag
Canola	Land preparation	1 bag
Sunflower	Germination; early growth	1 bag; 1 bag
Sesame	Germination; early growth	1 bag; ½ bag

VAN's published crop nutrition plans, 2025 set, as on van.com.pk/crops. 1 bag = 25 kg. The method for each row is printed in the crop's plan PDF.

The rate and stage from each crop's VAN plan.

Crop	Stage	Vital Urea per acre
Onion	Early growth (20 days after transplanting)	1 bag
Garlic	Land preparation; germination	1 bag; 1 bag
Tomato and chili	Land preparation or transplanting	1 bag (chili: 1 more bag at germination)
Watermelon	Land preparation; germination	1 bag; 1 bag
Strawberry	Land preparation	1 bag
Turmeric	Germination	1 bag
Chickpea, lentil, mungbean and mash, soybean	Land preparation	½ bag
Banana, year 1	Land preparation; germination; grand growth	1 bag at each stage
Banana, year 2	Land preparation; early growth	1 bag; 1 bag
Citrus	Land preparation; germination	2 bags; 1 bag

Date palm (per acre, by orchard age): 2.4 bags at years 1 to 2, 3.8 at 3 to 4, 7.5 at 5 to 6, 8.8 at 7 to 8, 11.3 at year 9 and over. Mango (per plant): 2 kg at land preparation, 0.84 kg at grand growth.

VAN's published crop nutrition plans, 2025 set, as on van.com.pk/crops. 1 bag = 25 kg. The method for each row is printed in the crop's plan PDF.

What to tell the farmer at the counter.

Do

- 1 Broadcast it, drill it at sowing, or side-dress it.
- 2 Apply it at the earliest stage in the crop plan.
- 3 Irrigate after applying, or apply before expected rain.
- 4 Take the rate from the crop's VAN plan.
- 5 Keep bags sealed, dry and off the floor.
- 6 Send the batch number if the farmer wants the certificate.

Do not

- 1 Do not fertigate or dissolve it in water.
- 2 Do not leave it on dry ground and wait weeks for water.
- 3 Do not crush or grind the granules.
- 4 Do not promise a yield figure. VAN publishes none for Vital Urea.
- 5 Do not say his soil is short of sulfur. Say it has never been tested.
- 6 Do not leave an open bag in the sun or the rain.

Not classified as hazardous under GHS. No signal word.

From VAN's safety data sheet, SDS-VAN-001, Revision 3, July 2026.

Storage

Cool, dry, well-ventilated place. Keep bags sealed and off the floor. Protect from moisture and direct sunlight.

Handling

Good industrial hygiene. Gloves and safety goggles; respiratory protection where dust is made. Wash hands after handling. Not for human or animal consumption. Keep away from children.

First aid

Eyes: rinse with water for several minutes. Skin: wash with soap and water. Swallowed: rinse mouth, do not induce vomiting, get medical attention. Inhaled: move to fresh air.

Spills and fire

Collect spillage into labelled containers; keep it out of drains and watercourses. Not flammable; use extinguishing media suited to the surrounding fire.

Transport

Not classified as a dangerous good for road, rail, sea or air.

Stability

Stable under normal storage and use. Keep away from strong acids and bases.

Helpline 0300 5003041. The full SDS is at van.com.pk/sds/VAN-SDS-Vital-Urea.pdf.

What has been tested, and how it was tested.

No yield percentage is claimed for Vital Urea. These are the 2 records VAN reports.

Published trial • maize, 2024

A trial by a fertilizer company on its own research farm, peer-reviewed and published. A randomised complete block design with 3 replications compared nitrogen programmes. 2 stage-timed applications of sulfur coated urea, at the 4 to 6 leaf stage and 20 days later at 8 to 12 leaves, did better across the growth and yield measures than a 5-application programme of urea followed by calcium ammonium nitrate.

Umair et al. (2025), Planta Animalia 4(3), 129 to 135, doi:10.71454/PA.004.03.0126. VAN cites the finding, not a headline number, because the paper reports its yield inconsistently between its table and its discussion.

Field programme • maize, Khaliqabad, 2023

With Rafhan Maize Products. 5 grower sites. One 25 kg bag of Vital Urea against one 50 kg bag of plain urea. Growth was comparable, with better uniformity of plant height, grain formation and milk-line development observed. Cob size and length showed no difference.

An observational field programme: not replicated, not randomised, and no independent yield measurement.

SOURCES

Where every figure in this deck came from.

Soil	Punjab soil testing programme, 36 district workbooks, 770,160 samples, c. 2016 to 2018, as published on van.com.pk.
Fertilizer use	NFDC Fertilizer Review 2024-25. Lassaletta, L., et al. (2014), Environmental Research Letters 9:105011.
Sulfur, India	Shukla, A. K., et al. (2021), Scientific Reports 11:19760, doi:10.1038/s41598-021-99040-2. Counter-evidence: Ahmad et al. (2003), Asian Journal of Plant Sciences 2(5).
Product	PS 217-2023; PSQCA licence CM/L-4126/2025; Pakistan Patent No. 144684; VAN SDS-VAN-001 Rev 3 (July 2026).
Release	VAN's own field observation of release start and end. Not a laboratory release curve.
Fine sulfur	Bremer, E. (2022), Crops & Soils 56(1), doi:10.1002/crso.20241. Janzen & Bettany (1987), Can. J. Soil Sci. 67(3), doi:10.4141/cjss87-057. Mattiello, E. M., et al. (2017), J. Agric. Food Chem. 65(6), doi:10.1021/acs.jafc.6b04586.
N and S	Yu, Z., et al. (2018), Scientific Reports 8:2499, doi:10.1038/s41598-018-20935-8. Yu, Z., et al. (2021), Communications Biology 4, 945, doi:10.1038/s42003-021-02458-7. Arata, A. F., et al. (2017), Crop & Pasture Science 68(3), 202 to 212, doi:10.1071/CP16330. Randall, Spencer & Freney (1981), Aust. J. Agric. Res. 32(2), 203 to 212, doi:10.1071/AR9810203. Grzebisz, W., et al. (2022), Agronomy 13(1), 66, doi:10.3390/agronomy13010066.
Removal	FAO (2006), Fertilizer and Plant Nutrition Bulletin 16. IRRI rice fact sheet (2003). Shukla & Lal (2004), Indian Journal of Agronomy 49(1).
Rates	VAN's published crop nutrition plans, 2025 set, van.com.pk/crops.
Evidence	Umair, A., et al. (2025), Planta Animalia 4(3), doi:10.71454/PA.004.03.0126. Khaliqabad field programme with Rafhan Maize Products, 2023.

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